

Jung Chang

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EXPERIENCE

Software Engineer Intern

Jan 2026 – April 2026

AGINT

Remote

- Designed and deployed AI agents using LangChain, LangGraph, and Cloudflare Workers, reducing manual workflow time by 60%.
- Optimized an intent-routing/classification system across 500+ test queries through iterative testing and refinement, improving accuracy by 18%.
- Integrated MCP (Model Context Protocol) and A2A agent endpoints via Streamable HTTP, resolving 10+ protocol-level issues and improving inter-agent reliability.
- Collaborated on backend API design and orchestration logic with senior engineers, reducing inter-service latency by 25%.
- Built a data pipeline to process 10,480 agent records from GoDaddy ANS, generating 152 validated manifests and increasing coverage 3x.

Front Desk / IT Support

Aug 2023 – May 2026

Progressive Urgent & Primary Care

Annapondale, VA

- Provided IT support by troubleshooting workstations, printers, and network connectivity for staff across the clinic.
- Digitized patient records using a cloud-based document system, reducing retrieval time by 40%.
- Optimized front-desk scheduling workflows using Microsoft 365 tools, contributing to a 20% reduction in appointment errors and improved staff efficiency.

PROJECTS

Algorithm Visualizer | *JavaScript, HTML5, CSS3, SVG, DOM API*

- Built an interactive web platform visualizing 20+ algorithms across 5 categories (sorting, graphs, trees, dynamic programming), enabling step-by-step execution and analysis.
- Designed a modular component architecture that reduced code duplication by 40% and simplified adding new algorithms.
- Implemented real-time animation rendering with efficient DOM updates and live metric tracking (time/space complexity, comparisons, swaps).

Automatic License Plate Recognition System | *Python, OpenCV, YOLOv8, EasyOCR, Pandas*

- Built an end-to-end ALPR pipeline using YOLOv8 and EasyOCR for automated license plate detection and recognition.
- Achieved 85.3% precision and 70.8% IoU on a 433-image dataset, outperforming baseline models by 20%+.
- Improved OCR accuracy in low-light conditions using CLAHE-based image preprocessing, and built a Pandas-based evaluation framework to analyze accuracy and false positives.

Calorie Tracker Pro | *JavaScript, Node.js, Express.js, REST APIs*

- Built a full-stack calorie tracking app with real-time search across 350K+ food items via USDA API integration.
- Implemented a secure Node.js/Express proxy layer to protect API keys and improve request reliability.
- Designed a responsive UI with optimized rendering, reducing page load time by 30%, and deployed on AWS Amplify with CI/CD pipelines.

Stock Research Tool (Web Application) | *Python, Flask, Pandas, Matplotlib, Plotly, SEC EDGAR XBRL API*

- Developed a full-stack web application that retrieves and processes financial data, with robust error handling for missing or incomplete records.
- Applied structured data validation and SQL-style querying with Pandas to derive key financial metrics.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript, HTML/CSS, SQL, MATLAB

Frameworks & Tools: React.js, Node.js, Express.js, Flask, LangChain, LangGraph

AI & Data: Pandas, NumPy, OpenCV, YOLOv8, EasyOCR

Platforms: Docker, AWS (Amplify), Cloudflare Workers, Git, Bash, Jupyter Notebook

Concepts: REST APIs, MCP/A2A protocols, Agile/Scrum, CI/CD, full-stack development

Certifications: CompTIA A+, CompTIA Network+ (In Progress)

EDUCATION

George Mason University

Fairfax, VA

Bachelor of Science in Computer Science

May 2026

- Relevant Coursework: Algorithms & Data Structures, Database Systems, Operating Systems, Computer Vision, Mobile Development, Object-Oriented Programming, Secure Programming & Systems.